Simulating a Fly's Biological Visual Network on FPGA

THE PROJECT

Biological neural circuits in hardware

We simulate how fruit fly visual circuits process brightness and motion, exposing the model's internal neural activity. We explore whether a circuit model grounded in biological neuron types and connectivity can run in full on a small chip.

Simulating biological vision on a chip

We map the pretrained FlyVis biological visual circuit model, with 45,669 neurons and 1.51 million connections, onto a Zynq-7020 FPGA. Camera images drive recurrent updates, while a PC displays neural activity and optical flow.

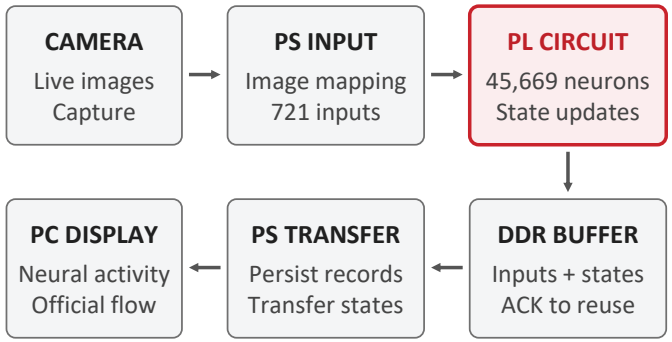
A prototype for future neural hardware

FPGA reconfigurability provides a working neural hardware prototype whose internal states can be observed and checked step by step. It supports exploration of low-energy neural chips and future brain-computer interfaces.

From a model to a working system

Camera, FPGA and PC operate together.
Source, firmware and launch tools are supplied.
Deployment on a teammate's PC is complete.

SYSTEM: BIOLOGICAL VISION IN HARDWARE



INNOVATION: BIOLOGICAL NEURAL CIRCUITS

WHY FPGA?

Toward low-energy neural chips
Reconfigurable circuits simulate biological vision.
A testable prototype for future neural chips.

Controlled computation and state updates
Observe and verify the complete neural state.

VISIBLE AND VERIFIABLE

Save every input and complete neural state.
Check every step with a software reference.

CREATIVE DESIGN

45,669
model neurons running as one network
All 1,513,231 connections preserved

AMD



AMD Zynq-7020 · OV5640

RESULT

ON THE BOARD: 1,000+ STEPS

Validation	Result
Continuous run	**1,014 steps**
State words checked	**46,308,366**
Bit mismatches	**0**

Independent check of saved board results.

ENERGY PER STEP

Platform and method	mJ/step
RTX 5060: measured	**166.71**
FPGA chip: estimated	**39.36**

OUTLOOK: TOWARD DIGITAL LIFE
Build verifiable neural hardware for future brain-computer interfaces and digital life that can sense, learn and interact.

GPU: NVML, including device memory.
FPGA: Vivado vectorless chip estimate.
1.788 W × 22.014 ms; PS, PL and static.

AMD